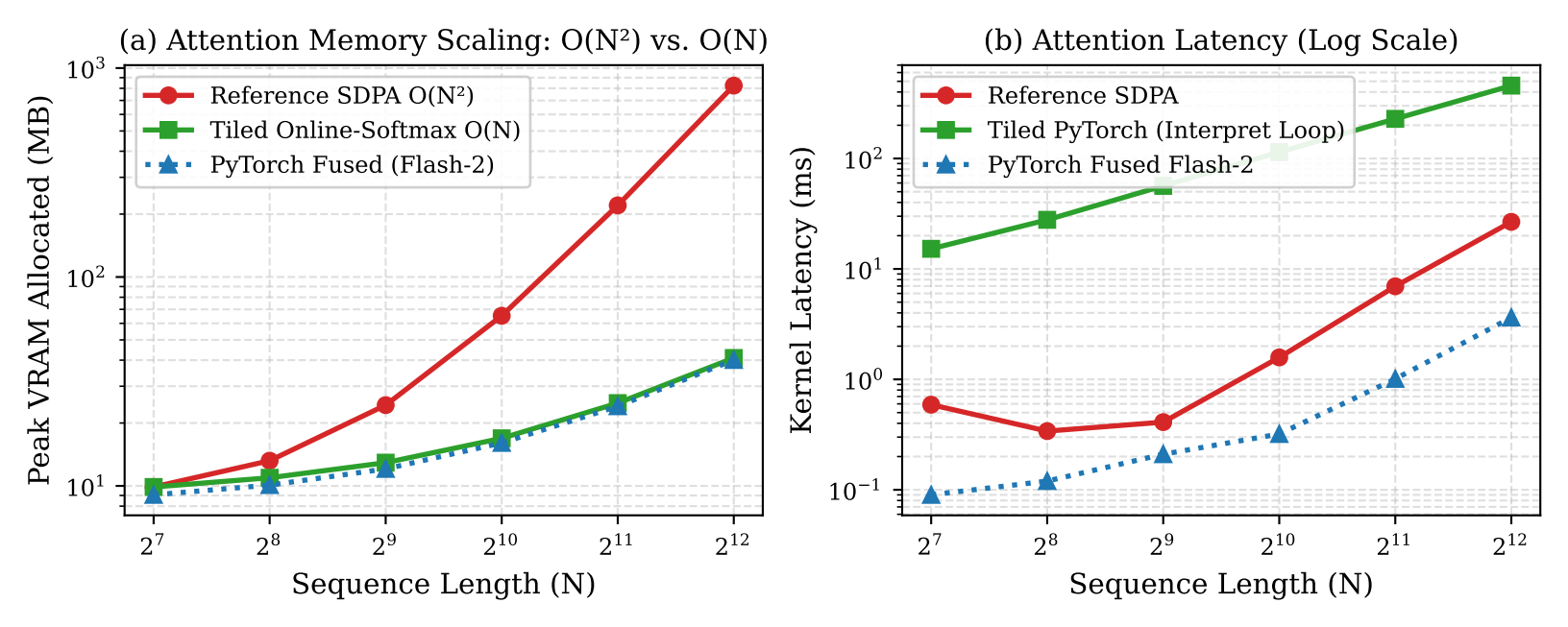
(a) Attention Memory Scaling: O(N²) vs. O(N)
Peak VRAM Allocated (MB)
Sequence Length (N)
Reference SDPA O(N²)
Tiled Online-Softmax O(N)
PyTorch Fused (Flash-2)
(b) Attention Latency (Log Scale)
Kernel Latency (ms)
Sequence Length (N)
Reference SDPA
Tiled PyTorch (Interpret Loop)
PyTorch Fused Flash-2